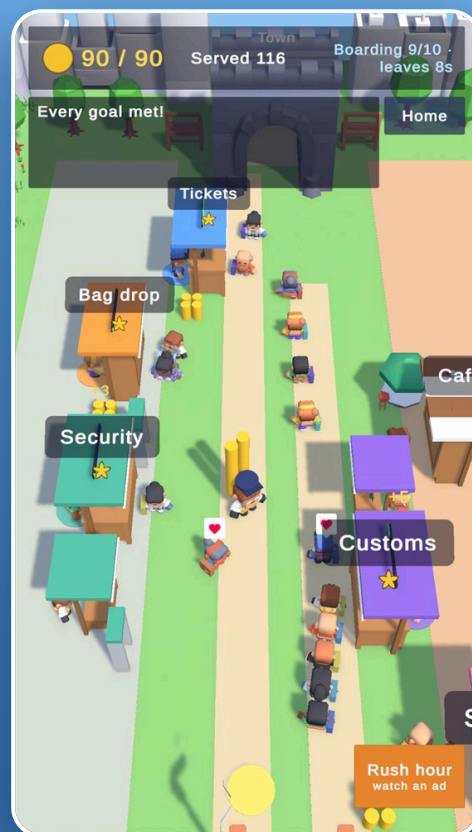
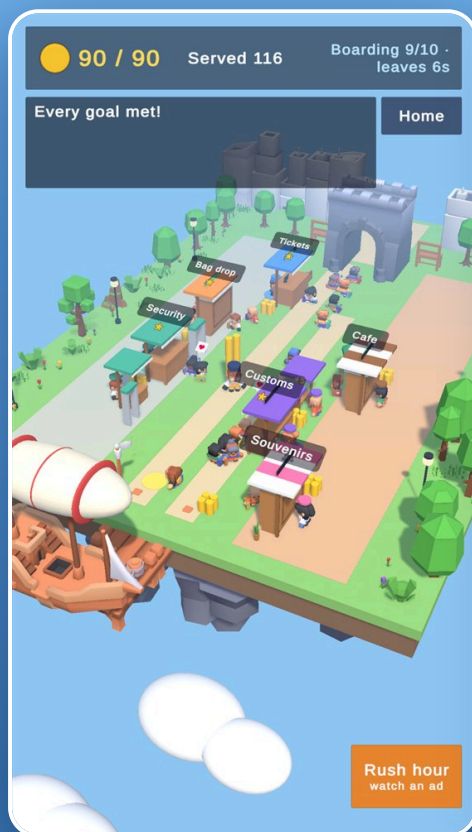


Cloudhaven Skyport

An arcade idle management game. Run an airship port on a floating island: serve travellers, carry the coins, build it up.



UMUT EREN DEMIRCI

Game design, systems, economy and balancing, UX, art and audio direction

September 2026 · Unity 6 · portrait mobile and web

The game in one screen

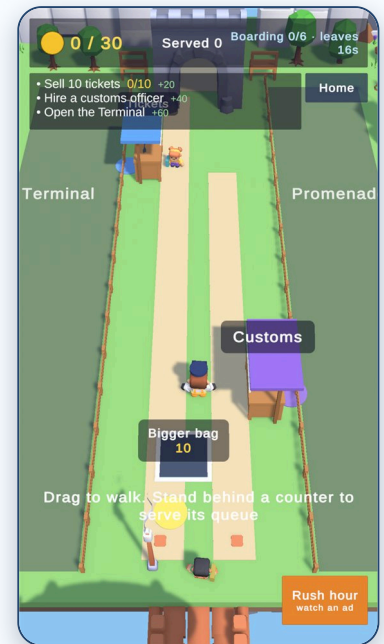
A floating island seen from above. Airships dock at the south end; the town gate is at the north. In between, a chain of counters the player builds, staffs and upgrades.

The pitch

Travellers arrive by airship and leave by airship. Arrivals go through customs to the town; departures come from the town through tickets, bag drop and security, may stop at the café or the souvenir stall, and board. Each counter serves only while someone stands behind it. The player is the harbourmaster: he serves whichever counter he stands at, picks up the coins it earns, and spends them by walking onto build pads.

Pillars

- **Walking is routing.** The player's time is the scarce resource; the decision is where to stand next. There are no chores that exist only to create friction.
- **Always something to buy.** A new pad every one to five minutes (the longest gap for a new player is 4:54).
- **Ads are opt-in boosts.** Rewarded ads double offline coins or start a rush hour; no upgrade is behind an ad.
- **Readable at a glance.** Queues, moods, coins and upgrades all show on the island itself.



The first minute

Audience

Arcade idle players (Airport Master, My Perfect Hotel): one thumb, portrait, sessions of 10 to 25 minutes, a clear sense of growth. The airship setting is new to the genre and gives each island its own look and twist.

References

GAME	WHAT IT DOES WELL	WHAT CLOUDHAVEN TAKES
Airport Master (Panteon)	An airport as a chain of stations, staff, shops; 10M+ downloads	The setting and the station chain, without chores (no sleeping staff) and without ads gating upgrades
My Perfect Hotel, Burger Please	Bright low-poly, an animation on every task, near 50% D1 (Udonis)	Chunky pastel low-poly, every action animated, a 50-minute first island
Mini Mart, Burger Please creatives	The stack on the character's back sells the game in an ad	The coin stack on the harbourmaster's back is the star of the ad
Idle tycoons (MindStudios analysis)	Capped offline income brings players back; a rewarded ad doubles it	Offline earnings capped at two hours, landing on the dock, doubled by an optional ad

Serve, pick up, spend

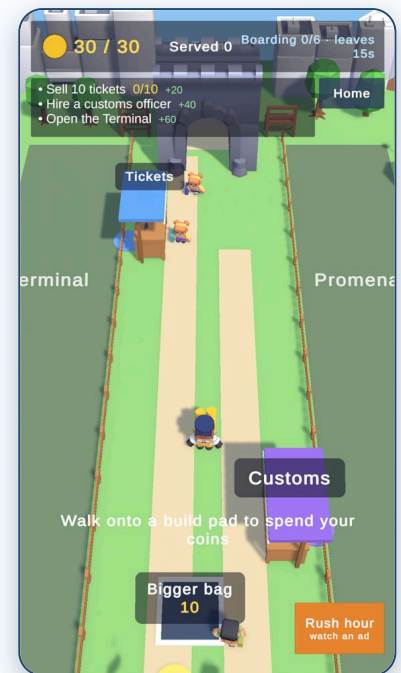
1. **Serve.** Stand behind a counter with a queue: a bar fills over its service time and the fare lands on the counter as coins. Served by the harbourmaster himself, a traveller tips 50%.
2. **Pick up.** Walk over the coin pile: coins arc onto the stack on his back (30 at the start, up to 90).
3. **Spend.** Stand on a build pad: coins pour in until it's paid, then the counter, clerk or zone appears.

Staff take over a counter at 45% of the harbourmaster's pace, so hiring frees him but never replaces him: the best income always comes from being where the queue is longest.

The counters

COUNTER	FLOW	SERVES IN	FARE	OPENS
● Customs	Arrivals	1.4 s	5	At the start
● Tickets	Departures	1.1 s	4	At the start
● Bag drop	Departures	1.2 s	3	Terminal (220)
● Security	Departures	1.5 s	3	Terminal (150)
● Café	Departures, 45% stop	2.2 s	10	Promenade (500)
● Souvenirs	Departures, 45% stop	2.6 s	15	Promenade (750)

Why the shops pay most: the café and the souvenir stall only see the 45% of departures who stop, and serve slowly, but they have their own shopkeeper and pay 10 and 15. They are the Promenade's reward: the first coins that arrive without the harbourmaster.



A full bag on his back

Three zones, 22 pads, one chain

The Harbour is open from the start; the Terminal and the Promenade are fenced off, shaded, and named, so the next goal is always in view.

● The Harbour

The dock, customs, the ticket desk and the gate. Two lanes: departures down one side, arrivals up the other. Teaches the loop in the first two minutes.

● The Terminal

Opens at about 4 minutes (target 3–7). Security and bag drop complete the departure chain; staff and speed upgrades follow.

● The Promenade

Opens at about 29 minutes (target 15–30). The café and the souvenir stall, then demand upgrades: bigger airships, more flights, the Grand Liner dock.

Every pad, in the order a new player buys them

Each pad appears when the one before it in the chain is bought. Times are the simulator's medians.

PAD	COST	NEW	EXPERT
Bigger bag	10	0:23	0:14
Customs officer	280	2:01	1:27
Open the Terminal	300	3:56	3:45
Security	150	4:46	4:31
Bag drop	220	6:44	6:10
Ticket clerk	300	9:47	9:08
Faster customs	250	11:58	11:09
More flights	450	15:55	15:08
Guard	400	18:22	17:27
Porter's cart	300	20:27	19:08
Porter	550	23:52	21:52

PAD	COST	NEW	EXPERT
Faster bag drop	450	25:41	24:00
Open the Promenade	800	29:19	27:35
Cafe	500	29:21	27:37
Faster security	600	31:10	29:14
Souvenir stall	750	34:10	32:10
Bigger airships	1300	38:41	36:15
Faster tickets	700	40:20	37:54
Second customs desk	900	42:12	39:18
Second security lane	1100	44:48	41:16
Even more flights	1800	48:12	43:54
Grand Liner dock	1800	53:05	47:16

Four kinds of pad. *Demand* (more flights, bigger airships, the Liner) and *staff* pay back in 5 to 17 minutes. *Capacity* (faster counters, second counters) barely shows on its own but keeps queues short as demand grows. *New counters and zones* can dip income for a few minutes, because each is one more queue until it's staffed; they open the next purchases.

Queues you can read

Flow

- Arrivals step off the airship, queue at customs and walk to the town gate.
- Departures come from the gate through each built counter in order, then each shop tempts them on its own (45% each), then they join the boarding queue at the head of the dock.
- Queues trail back toward where travellers come from; a long queue snakes back and forth beside the counter, and a rule test checks no queue slot on any island ever leaves the island.

Patience

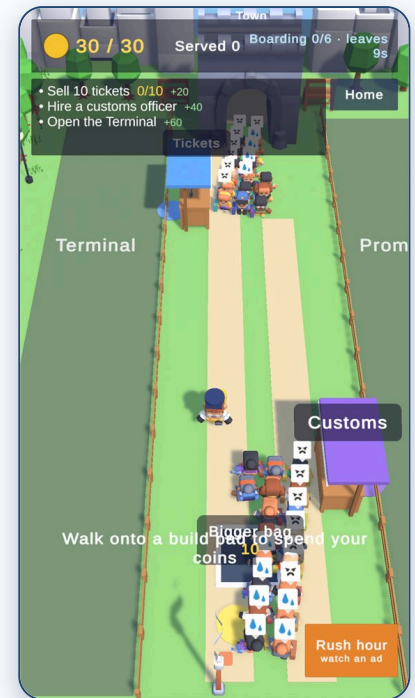
A traveller waits up to 40 seconds in queues. After that they turn grumpy: they pay half and skip the shops. A new player's island runs at about 10% grumpy; the tuning story in section 6 shows it at 46% when capacity lagged.

Moods, shown

A speech bubble over each traveller, most urgent first: an angry face (grumpy), a heart (just served), sweat drops (half their patience gone) and a star (a VIP). The player can read a failing counter from across the island.

Twists

- VIPs (Lantern Reach, 15% of travellers) pay three times the fare but have half the patience.
- Grand Liners: every fifth airship carries twice the travellers, with a toast, a gold envelope and its own music while docked.



An unattended customs queue

Supply is the pacing lever

The island's income is capped by how many travellers the airships bring, so the timetable, not the counters, decides how fast an island can grow.

RULE	VALUE	WHY
Airship interval	40 s	One ship per loop around the island at a walking pace
Approach, then docked	3 s, then 16 s	Arrivals step off every 0.6 s; departures board every 0.5 s
Travellers each way	6 (10 with bigger airships)	Demand grows with purchases, not with time
Full-ship bonus	3 coins a seat	Rewards keeping the departure chain moving; lands on the dock
Grand Liner	every 5th ship (every 3rd with its dock), ×2 travellers	A rush the player can see coming
More flights	6 s off the interval, twice (40 → 28 s)	The strongest late purchase
Storms (Galewind)	every 4th ship 12 s late, ×1.5 travellers	Bunched crowds reward spare capacity



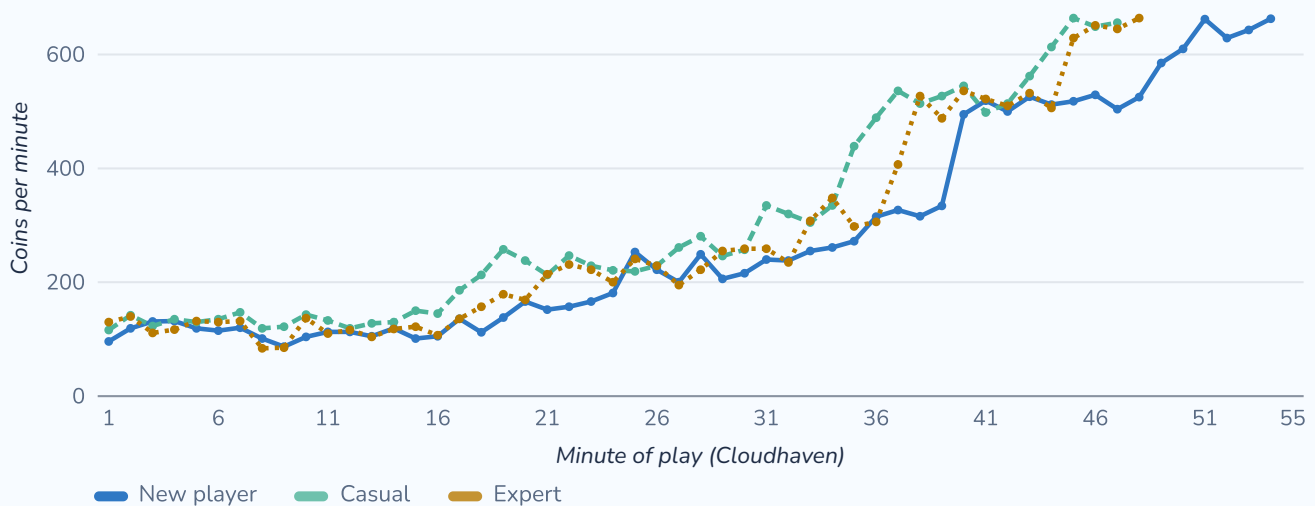
A Grand Liner docked



A storm ship on Galewind Isle

Bots play the real rules

The economy simulator compiles the game's own rules ([Cloudhaven.Core](#)) and plays each island 40 times with three stand-in players: a new player who wanders, pauses and buys late, a casual, and an expert who always serves the busiest counter. Its CSVs feed the balance workbook.



MOMENT	TARGET	NEW	CASUAL	EXPERT
First coin	0:00–0:10	0:07	0:05	0:05
First pad (Bigger bag)	0:00–0:30	0:23	0:18	0:14
First staff (customs officer)	2:00–3:00	2:01	1:43	1:27
The Terminal opens	3:00–7:00	3:56	3:41	3:45
The Promenade opens	15:00–30:00	29:19	25:21	27:35
Every pad bought	30:00–55:00	53:05	46:16	47:16
Island complete	40:00–60:00	53:05	46:16	47:16

How it got here

1. Everyone finished in 15 minutes: income was capped by the timetable, so skill didn't matter.
2. Demand upgrades (more flights, bigger airships, the Liner) and prices that climb along the chain: 52 minutes.
3. Slower staff made the harbourmaster matter, but demand outgrew the counters: 46% of travellers turned grumpy.
4. Capacity to match demand (faster counters, second counters, 40 s of patience): grumpy back to 10%.
5. Tips and independent shops: a hands-on player now finishes about six minutes sooner than a new one.

Three islands, one sky

ISLAND	MONEY	TWIST	NEW PLAYER	EXPERT
● Cloudhaven	×1	None: learn the ropes	53:05	47:15
● Galewind Isle	×2	Storms; the layout is mirrored	58:43	48:36
● Lantern Reach	×4	VIPs: 3× fare, ½ patience	51:01	42:08

Same bones, new problem. Every island has the same 22 pads and 10 goals, so the player knows the chain; the twist changes what matters. Prices, fares, rewards, carry capacity and pay-in speed scale with the island's multiplier, so a ×4 island is not four times longer (scaling prices alone made Galewind 75 minutes and Lantern unfinishable).

Idle earnings

Staffed islands keep earning while the player is away: a flow model of the staffed counters at staff pace, at 30% of the played rate, capped at two hours. If any built departure counter is unstaffed, departures earn nothing, so the player must staff the whole chain. The coins land on the dock; an optional ad doubles them.

Rush hour

An optional ad doubles every fare for 60 seconds.

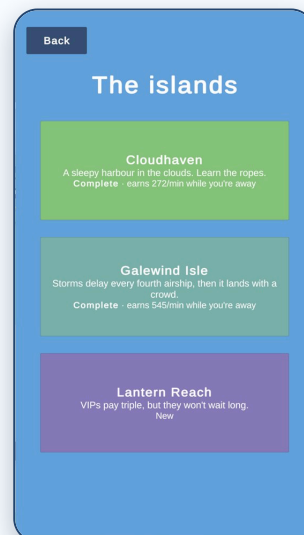
Saving

JSON every 10 seconds, on every purchase, and on pause and quit; written through a temp file. Restoring re-applies each purchase through the same code the game uses, and a test round-trips an island.

Daily event

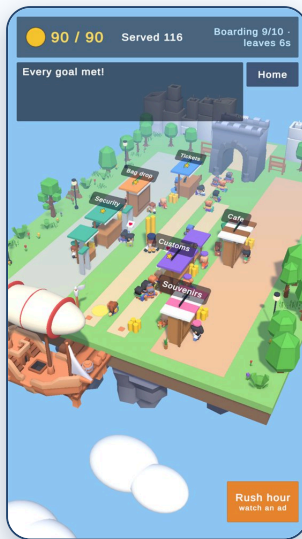
One per day, from a hash of the date: the same for everyone. All four are bonuses: a reason to open the app, never a penalty.

DAILY EVENT	EFFECT
Sky Festival	The café and souvenir stall charge double
Tailwind	Airships arrive 20% more often
Tourist Season	Two more travellers each way on every airship
Calm Skies	Patience ×1.5

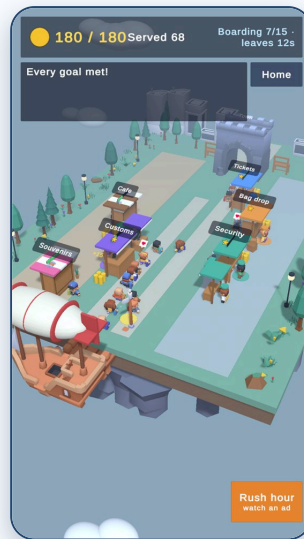


The islands map

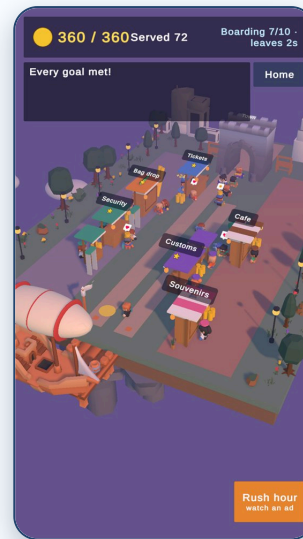
A floating island



Cloudhaven: spring



Galewind: windswept



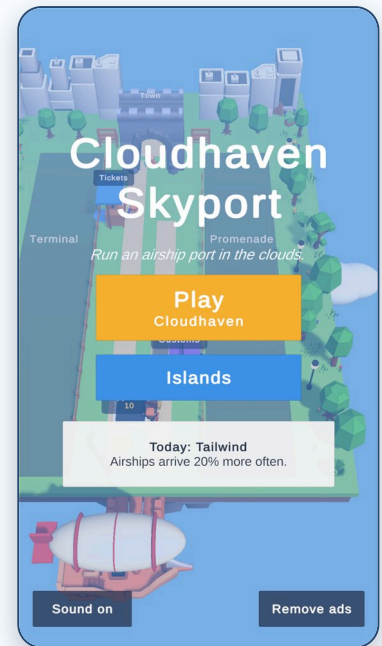
Lantern Reach: dusk

Every asset is CC0: six Kenney model kits (animated mini characters, pirate ships and docks, nature, furniture, food, a town), four Kenney sound packs and five OpenGameArt tracks, one per island plus the menus and the Grand Liner. Each island has its own palette, light, fog and music.

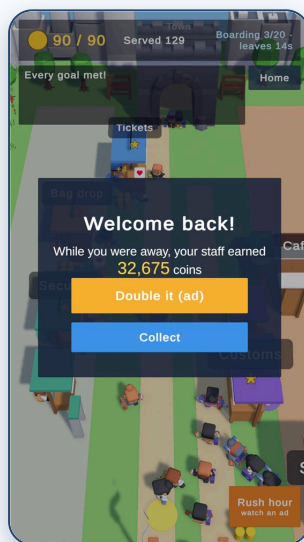
MOMENT	FEEDBACK
Picking up coins	Coins arc onto the stack; a clink rising in pitch as it fills; the HUD coin bumps
Walking	The stack leans against the motion
Buying	Confetti and a chime; a new counter lands with a bounce and dust; new staff pop in and nod
Serving	A floating +5 over the counter, a soft pluck, a heart over the traveller
A zone opens	The fence drops post by post, the floor changes, confetti, a camera shake, a jingle
Airships	A bell on approach; the Grand Liner's own music while docked; rain under storm clouds

Teach by doing, show on the island

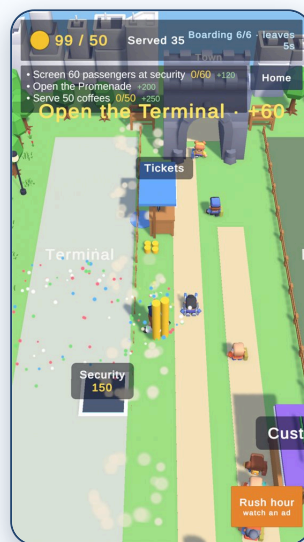
- Three hints, then silence. "Drag to walk, stand behind a counter", "Pick up the coins", "Walk onto a build pad". They stop at the first purchase.
- Goals, three at a time, each with a reward that lands in the bag; progress in gold, rewards in green.
- Floating joystick anywhere on the screen; a touch that starts on a button never moves the harbourmaster.
- Signs on plates: counter names and pad prices read on any ground; each speed upgrade adds a gold star.
- Second counters are real: a booth with its own clerk, placed by a rule that keeps it off queues and pads.
- The home screen shows the game: the current island plays itself behind a translucent menu.
- Portrait, safe area, one thumb; the web build keeps the phone's shape on desktop.



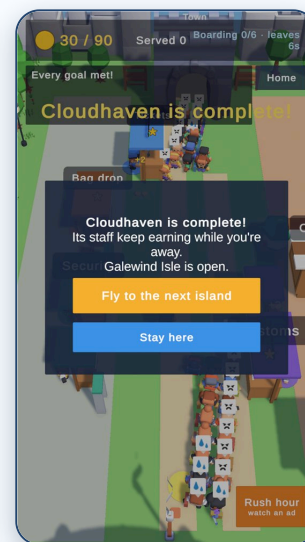
Home: the island behind the menu



Welcome back



The Terminal opens



Island complete

Rules in one place

Architecture

- `Cloudhaven.Core` : every rule in plain C#, no UnityEngine; the island ticks at 60 Hz and emits events.
- `Cloudhaven.Game` : MonoBehaviours only display, take input and animate.
- The same Core runs in the **economy simulator** (.NET, outside Unity) and in **32 EditMode tests**.
- The scene is built from code; art and sound sets are assets filled by an editor script.
- Ads, purchases and analytics go through `IAdService`, `IStoreService`, `IAalytics`; a placeholder grants everything and events are logged locally.

Builds

- Android APK: IL2CPP, ARM64, API 26+.
- WebGL for itch.io, with a phone-shaped page and home-screen icons.

Roadmap

NEXT	WHY
Playtest the mid-island plateau	Minutes 5–15 are the longest stretch without growth
Island 4 and a helper booster	Content depth; a rewarded helper who carries coins
Cosmetic airships	The first IAP beyond "remove ads"
Real ad and analytics SDKs	Drop-in behind the existing interfaces
Soft launch with the KPI plan	See the Ad and KPI Plan document

Risks

- Bots aren't people: real first sessions are slower; check the 53-minute island against real playtimes.
- Galewind's Promenade opens at 34:37, just past the 30-minute target: the storms' price.